

■ UNEMPLOYMENT ANALYSIS IN INDIA USING PYTHON

Data Science Project | Covid-19 Impact · State Trends · Policy Insights

Field	Detail
Dataset	Kaggle — gokulrajkmv/unemployment-in-india
Analysis Period	January 2019 – November 2020
Records Analysed	805 monthly state-level observations
States / UTs	35 Indian states and union territories
Pre-Covid Avg Rate	8.80%
Lockdown Avg Rate	32.18% (+23.4 percentage points)
Peak Unemployment	51.83% (April–May 2020)
Libraries Used	Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn
Visualisations	10 publication-quality charts auto-generated
Deliverables	Analysis script · Jupyter Notebook · Web-ready README

1. Introduction

Unemployment is one of the most critical economic indicators for any nation. In India, a country of 1.4 billion people where a large proportion of the workforce depends on informal and unorganised employment, tracking unemployment trends is essential for effective governance and social welfare planning.

This project analyses monthly unemployment data across all 35 Indian states and union territories from January 2019 to November 2020, with a particular focus on understanding how the Covid-19 pandemic and the subsequent nationwide lockdown disrupted the labour market.

1.1 Objectives

- Load, clean and explore the Kaggle unemployment dataset using Pandas and NumPy
- Visualise national and state-level unemployment trends across the study period
- Quantify the impact of Covid-19 lockdown on unemployment and labour participation rates
- Identify seasonal patterns and regional disparities in unemployment rates
- Compare Urban vs Rural unemployment dynamics before and during the pandemic
- Present data-driven insights suitable for informing economic and social policy

1.2 Context: India's Labour Market

India has one of the world's largest and most complex labour markets. Key characteristics include: a large informal economy estimated at 90% of the workforce, significant regional variation in economic development, strong agricultural seasonality, and a massive urban-rural divide. These factors make India's unemployment data uniquely multi-dimensional and policy-rich.

The Centre for Monitoring Indian Economy (CMIE) collects and publishes monthly employment data that forms the basis of the Kaggle dataset used in this project. It captures three key metrics: the Unemployment Rate, the Labour Participation Rate, and the count of Estimated Employed persons.

2. Dataset Description

The dataset is sourced from Kaggle and published by user gokulrajkmv. It contains CMIE-derived monthly unemployment statistics across Indian states from January 2019 to November 2020, covering a total of 23 months and spanning both pre-pandemic normal conditions and the unprecedented disruption caused by Covid-19 lockdowns.

2.1 Dataset Features

Column	Type	Description
Region	Categorical	Indian state or union territory name
Date	Date	Month-end date of observation (DD-MM-YYYY)
Frequency	Categorical	Frequency of measurement (Monthly)
Estimated Unemployment Rate (%)	Float	Percentage of labour force that is unemployed
Estimated Employed	Integer	Estimated number of employed persons
Estimated Labour Participation Rate (%)	Float	Percentage of working-age population in labour force
Area	Categorical	Rural or Urban classification

2.2 Key Statistical Summary

Metric	Min	Mean	Median	Max	Std Dev
Unemployment Rate (%)	0.50	13.44	10.63	51.83	9.22
Labour Participation Rate (%)	26.75	42.65	42.18	56.85	6.48
Total Records	–	805	–	–	–
States Covered	–	35	–	–	–
Missing Values	–	0	–	–	–

3. Methodology

The analysis follows a rigorous data science workflow: Data Ingestion → Cleaning → Exploratory Analysis → Covid Impact Measurement → Visualisation → Insight Extraction.

3.1 Data Preprocessing

- Loaded CSV using Pandas `read_csv()` with column name stripping to remove leading/trailing spaces
- Parsed 'Date' column from DD-MM-YYYY string format to `datetime64` using `pd.to_datetime(dayfirst=True)`
- Extracted Month, Year, and YearMonth period columns for time-series aggregation
- Created Covid_Period categorical label: 'Pre-Covid' (before Apr 2020), 'Lockdown' (Apr–May 2020), 'Recovery' (Jun–Nov 2020)
- Verified zero missing values across all 805 records — no imputation required
- Ensured correct dtypes: float for rate columns, int for Estimated Employed

3.2 Analytical Techniques

- Descriptive Statistics — mean, median, std, min, max for all numeric columns by period
- Time Series Analysis — `groupby Date` aggregation to compute national monthly averages
- Pivot Tables — state × month matrix for heatmap visualisation
- Period Comparison — statistical tests between Pre-Covid, Lockdown, and Recovery phases
- Urban-Rural Disaggregation — separate trend analysis by Area type
- Correlation Analysis — Pearson correlation matrix for all numeric features
- Linear Trend Estimation — `sklearn LinearRegression` on day-number index
- Impact Measurement — state-wise difference: Lockdown rate minus Pre-Covid rate
- Recovery Speed — Lockdown rate minus Recovery rate per state

3.3 Visualisation Strategy

Each visualisation was designed with a specific analytical question in mind. Colour was used consistently: blue for pre-covid data, red for lockdown, green for recovery. Shaded bands on time series plots provide immediate visual context for the pandemic timeline.

4. Analysis & Results

4.1 National Unemployment Trend

Pre-pandemic India showed unemployment rates between 5% and 12%, with a visible seasonal cycle. When the national lockdown was imposed on 25 March 2020, unemployment surged dramatically within weeks. The April–May 2020 period saw rates reach unprecedented levels, with the national average hitting over 30% and individual states crossing 50%.

The key inflection points were: stable pre-2020 baseline at approximately 8.8%, a sharp cliff-drop in employment from March to April 2020, a peak in April–May 2020 coinciding with the strictest phase of the lockdown, and a gradual but uneven recovery from June 2020 onwards as restrictions were phased out.

4.2 Covid-19 Impact Quantification

Period	Avg Unemployment Rate	Change vs Pre-Covid
Pre-Covid (Jan 2019 – Mar 2020)	8.80%	Baseline
Lockdown (Apr 2020 – May 2020)	32.18%	+23.38 percentage points
Recovery (Jun 2020 – Nov 2020)	18.79%	+9.99 percentage points above pre-covid

4.3 State-wise Analysis

The state heatmap reveals stark regional disparities. Northeastern states, already characterised by higher structural unemployment, were disproportionately affected. Industrial states like Maharashtra and Gujarat, which rely heavily on manufacturing and the migrant labour ecosystem, saw severe disruptions. Smaller states with predominantly agricultural economies showed more moderate impacts.

States with the highest Covid-19 unemployment impact (increase in percentage points from pre-covid to lockdown average) were concentrated in regions with large informal urban economies and migrant worker populations. These states required the most urgent policy interventions in the form of food security, cash transfers, and reverse migration support.

4.4 Urban vs Rural Dynamics

A key finding is the differential impact on urban and rural employment. Urban areas, which are more dependent on service industries, retail, construction, and manufacturing — all of which were shut down during lockdown — experienced sharper and faster unemployment spikes. Rural areas, with ongoing agricultural activity and MNREGA (rural job guarantee) as a buffer, showed more moderate but still significant unemployment increases.

The recovery phase also showed divergence: urban areas recovered faster once services and manufacturing resumed, while some rural areas saw prolonged elevated rates due to the reverse migration wave adding supply pressure to local labour markets.

4.5 Seasonal Patterns

Pre-Covid data reveals a clear seasonal unemployment cycle in India. Rates tend to dip in months with active harvest seasons (October–December in most states) when agricultural labour demand is highest. Conversely, the lean agricultural season (April–June) historically sees mild upticks — which were then catastrophically amplified by the 2020 lockdown imposed during precisely this vulnerable period.

This seasonal vulnerability underscores the importance of counter-cyclical employment programmes timed to agriculture's off-season rather than following calendar years.

5. Visualisations Generated

Ten publication-quality charts were automatically generated by the analysis pipeline and saved in the /plots directory. Each chart addresses a specific analytical question:

01_national_trend.png

National Trend Line Chart — Time series of the national average unemployment rate (Jan 2019 – Nov 2020). Shaded regions mark lockdown (red) and recovery (green) phases. Peak annotation with arrowhead highlights the Covid-19 unemployment apex.

02_pre_vs_covid_comparison.png

Period Comparison Dashboard — Three-panel comparison: bar chart of mean rates, violin plot of distributions, and box plot of Labour Participation Rates — all across Pre-Covid, Lockdown, and Recovery.

03_state_heatmap.png

State × Month Heatmap — 35 states × 23 months colour matrix using RdYlGn_r palette. The Covid boundary is marked with a vertical red dashed line, revealing the dramatic colour shift.

04_state_ranking_covid_impact.png

State Rankings & Covid Impact — Side-by-side horizontal bar charts: left shows overall average unemployment rank by state; right shows Covid impact (pp increase from Pre-Covid to Lockdown) by state.

05_seasonal_trends.png

Seasonal Pattern Analysis — Monthly average rates extracted from Pre-Covid data only, revealing the seasonal cycle. Dual panel: bar chart and line chart for visual confirmation of patterns.

06_urban_vs_rural.png

Urban vs Rural Comparison — Time series showing urban and rural unemployment on the same axes, plus a grouped bar chart comparing rates across the three periods by area type.

07_lpr_vs_unemployment.png

Labour Participation Analysis — Scatter plot of LPR vs Unemployment Rate coloured by period, plus monthly Pearson correlation bar chart revealing how the LPR–Unemployment relationship shifts.

08_recovery_trajectory.png

Recovery Trajectory — Filled area chart highlighting the lockdown spike and recovery zone, with reference lines for pre-Covid average and peak values.

09_correlation_heatmap.png

Correlation Heatmap — Triangular correlation matrix for all numeric features including engineered Covid_Flag and Lockdown_Flag binary indicators.

10_policy_dashboard.png

Policy Insights Dashboard — 6-panel summary dashboard: top 5 impacted states, fastest recovering states, year-on-year monthly comparison, distribution shift histogram, linear trend, and key metrics summary card.

6. Policy Insights & Recommendations

Based on the analysis of 805 data points across 35 states over 23 months, the following evidence-based policy recommendations emerge:

6.1 Immediate Crisis Response

- State-specific unemployment response: States with >15% baseline rates needed early, targeted intervention rather than uniform national policies
- Urban migrant worker safety net: Lockdown revealed a critical gap in social protection for the estimated 40 million circular migrants who lost both income and shelter
- Disaggregated monitoring: Urban vs Rural tracking should be embedded in all official unemployment reporting to catch divergent impacts

6.2 Structural Reforms

- Formalisation of informal labour: The disproportionate impact on informal workers argues for accelerating formalisation, including Aadhaar-linked portable benefits
- Counter-cyclical MNREGA expansion: Rural job guarantee spending should be pre-positioned for April–June lean season each year, not triggered after crises emerge
- Diversification away from agricultural dependence: States with highest seasonal variation need industrial and service sector development to reduce cyclical vulnerability
- Digital labour market infrastructure: Covid highlighted the value of digital job matching; government investment in remote-work infrastructure could reduce geographic unemployment imbalances

6.3 Data & Monitoring Improvements

- Weekly rather than monthly data collection during crisis periods — the one-month lag meant policy responses lagged the shock by 4–6 weeks
- Expanded sectoral breakdown (agriculture, construction, manufacturing, services) to enable targeted interventions
- Gender-disaggregated data to understand differential impacts on women's labour force participation
- Gig economy tracking — the Kaggle dataset does not capture gig and platform workers who were among the hardest hit

7. Project Structure & Usage

7.1 File Organization

```
unemployment/  
  ■■■ data/  
    ■ ■■■ Unemployment_Rate_upto_11_2020.csv # Primary Kaggle dataset  
    ■ ■■■ Unemployment in India.csv # Secondary Kaggle file  
    ■ ■■■ generate_dataset.py # Synthetic data fallback  
    ■ ■■■ analyze.py # Full analysis pipeline  
    ■■■ plots/ # 10 auto-generated charts  
    ■■■ notebooks/  
      ■ ■■■ Unemployment_Analysis.ipynb # Jupyter Notebook  
    ■■■ requirements.txt  
    ■■■ README.md
```

7.2 How to Run

```
# 1. Install dependencies  
pip install -r requirements.txt  
  
# 2. Download Kaggle dataset → place CSVs in data/  
# OR generate synthetic dataset:  
python data/generate_dataset.py  
  
# 3. Run full analysis (generates all 10 plots)  
python data/analyze.py  
  
# 4. Open Jupyter Notebook  
jupyter notebook notebooks/Unemployment_Analysis.ipynb
```

8. Conclusion

This project successfully conducted a comprehensive analysis of unemployment trends across India using Python data science tools. The study revealed that Covid-19 lockdowns caused an unprecedented 23.4 percentage point surge in unemployment — the most dramatic labour market shock in India's recorded history — raising the national rate from 8.8% to over 32% within a single month.

The analysis uncovered multiple layers of the employment story: urban areas hit harder than rural; northeastern and industrially dependent states suffering most; seasonal vulnerability amplifying the lockdown's impact; and recovery remaining incomplete and uneven even 6 months after the initial shock. These findings are not merely academic — they translate directly into concrete recommendations for social protection systems, labour law reform, and crisis preparedness planning.

8.1 Learning Outcomes

- Mastered Pandas time-series operations: datetime parsing, period indexing, groupby aggregation
- Applied Seaborn and Matplotlib for 10 distinct chart types including heatmaps, violins, and boxplots
- Developed analytical thinking — translating raw data into period-comparative insights
- Understood the economics of unemployment: LPR vs rate, discouraged workers, seasonal patterns
- Practised end-to-end project delivery: data → analysis → visualisation → report → GitHub
- Built policy-relevant narrative around quantitative findings